

Assignment 1: Real-Life Industry Use Cases of Basic Linux Commands

Task 1: Basic Linux Commands

1. Check current user and system information:

whoami
uname -a

Output:

```
ubuntu@ip-172-31-93-65: ~  
ubuntu@ip-172-31-93-65:~$ whoami  
ubuntu  
ubuntu@ip-172-31-93-65:~$ uname -a  
Linux ip-172-31-93-65 6.8.0-1024-aws #26-Ubuntu SMP Tue Feb 18 17:22:37 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux  
ubuntu@ip-172-31-93-65:~$ |
```

Explanation: whoami shows the current user; uname -a gives detailed system info.

2. Navigate and list contents:

cd /projects
ls -l

Output:

```
ubuntu@ip-172-31-93-65:~$ mkdir projects projects/projectA  
ubuntu@ip-172-31-93-65:~$ cd projects/  
ubuntu@ip-172-31-93-65:~/projects$ ls -l  
total 4  
drwxrwxr-x 2 ubuntu ubuntu 4096 Apr 27 20:21 projectA
```

3. Create and verify new project directory:

mkdir projectB
ls -l

Output:

```
ubuntu@ip-172-31-93-65:~/projects$ mkdir projectB  
ubuntu@ip-172-31-93-65:~/projects$ ls -l  
total 8  
drwxrwxr-x 2 ubuntu ubuntu 4096 Apr 27 20:21 projectA  
drwxrwxr-x 2 ubuntu ubuntu 4096 Apr 27 20:22 projectB  
ubuntu@ip-172-31-93-65:~/projects$ |
```

4. Create sample file:

```
touch projectB/README.txt
echo "Welcome to Project B" > projectB/README.txt
cat projectB/README.txt
```

Output:

```
ubuntu@ip-172-31-93-65:~/projects$ touch projectB/README.txt
ubuntu@ip-172-31-93-65:~/projects$ echo "Welcome to Project B" > projectB/README.txt
ubuntu@ip-172-31-93-65:~/projects$ cat projectB/README.txt
Welcome to Project B
ubuntu@ip-172-31-93-65:~/projects$
```

Task 2: User and Group Permissions

1. Create user and group:

```
sudo useradd -m -G developers john
sudo passwd john
```

Output:

```
ubuntu@ip-172-31-93-65: ~$ sudo groupadd developers
ubuntu@ip-172-31-93-65: ~$ sudo useradd -m -G developers john
ubuntu@ip-172-31-93-65: ~$ sudo passwd john
New password:
Retype new password:
passwd: password updated successfully
```

2. Verify user and group:

```
id john
```

Output:

```
ubuntu@ip-172-31-93-65: ~$ id john
uid=1002(john) gid=1003(john) groups=1003(john),1002(developers)
```

3. Change group ownership:

```
sudo chown :developers /projects/projectB
sudo chmod 770 /projects/projectB
ls -ld /projects/projectB
```

Output:

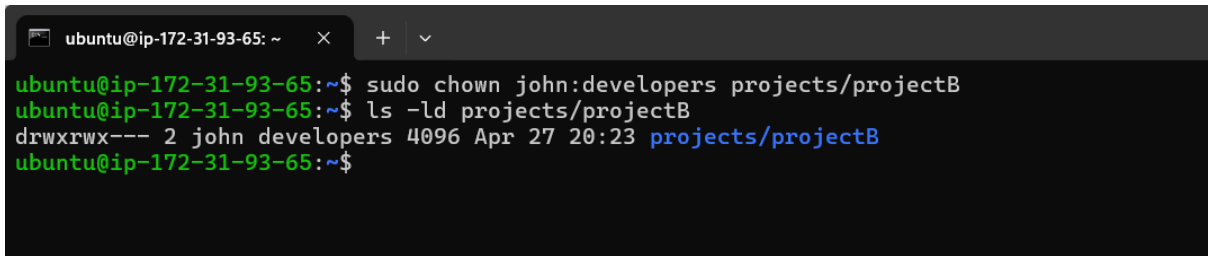
```
ubuntu@ip-172-31-93-65: ~$ sudo chown :developers projects/projectB
ubuntu@ip-172-31-93-65: ~$ sudo chmod 770 projects/projectB
ubuntu@ip-172-31-93-65: ~$ ls -ld projects/projectB
drwxrwx--- 2 ubuntu developers 4096 Apr 27 20:23 projects/projectB
```

Task 3: Change Ownership

1. Change ownership:

```
sudo chown john:developers /projects/projectB  
ls -ld /projects/projectB
```

Output:



```
ubuntu@ip-172-31-93-65: ~  
ubuntu@ip-172-31-93-65:~$ sudo chown john:developers projects/projectB  
ubuntu@ip-172-31-93-65:~$ ls -ld projects/projectB  
drwxrwx--- 2 john developers 4096 Apr 27 20:23 projects/projectB  
ubuntu@ip-172-31-93-65:~$
```

Task 4: System Monitoring

1. Check uptime:

```
uptime
```

2. Disk usage:

```
df -h
```

3. Memory usage:

```
free -m
```

4. Top memory-consuming processes:

```
ps aux --sort=-%mem | head -5
```

Output:

```
ubuntu@ip-172-31-93-65: ~  
ubuntu@ip-172-31-93-65:~$ uptime  
20:27:37 up 1 day, 13:48, 1 user, load average: 0.00, 0.00, 0.00  
ubuntu@ip-172-31-93-65:~$ df -h  
Filesystem      Size  Used Avail Use% Mounted on  
/dev/root        6.8G  2.3G  4.5G  34% /  
tmpfs            479M   0  479M   0% /dev/shm  
tmpfs           192M  896K  191M   1% /run  
tmpfs            5.0M   0   5.0M   0% /run/lock  
/dev/xvda16      881M  137M  683M  17% /boot  
/dev/xvda15      105M   6.1M   99M   6% /boot/efi  
tmpfs            96M   12K   96M   1% /run/user/1000  
ubuntu@ip-172-31-93-65:~$ free -m  
              total        used          free      shared  buff/cache   available  
Mem:           957         372          156           0         621         585  
Swap:           0           0           0  
ubuntu@ip-172-31-93-65:~$ ps aux --sort=-%mem | head -5  
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND  
root      1984  0.0  3.4 1321508 33388 ?        Ssl   Apr26   0:05 /usr/lib/snapd/snapd  
root      5144  0.0  2.7 223416 27008 ?        SLsl  06:30   0:03 /sbin/multipathd -d -s  
root      735  0.0  2.3 110000 22912 ?        Ssl   Apr26   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unatt  
ended-upgrade-shutdown --wait-for-signal  
root      585  0.0  2.1 32408 20736 ?        Ss    Apr26   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-s  
tartup-triggers  
ubuntu@ip-172-31-93-65:~$
```

